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Palaeomagnetism of Proterozoic mafic intrusions and host rocks of the Mount Isa Inlier, Australia: revisited

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Abstract

A palaeomagnetic study was carried out on Proterozoic dolerite dykes, plugs and some of their host rocks in the intrusion-rich Mount Isa Inlier for which only one other similar study has been reported (Duff and Embleton, 1976). After rejection of intrusions with unstable remanences and internally inconsistent directions, the results from eleven dykes and five plugs remain. These yield two main palaeomagnetic directions: a direction with both polarities and moderate northerly upward (southerly downward) inclination, obtained from metamorphosed dykes and plugs; and a near-vertical downward direction, obtained from unmetamorphosed dykes. The first appears to be associated with a period of regional metamorphism in the Mount Isa Inlier, estimated at ~1620–1500 Ma, and having the last major metamorphic peak near 1554 Ma. A moderate directional dispersion suggests that this magnetization post-dates periods of major deformation of the region, estimated to have ended at around 1550 Ma. Thus the likely age of the magnetization is 1550–1500 Ma or younger, depending on the rate of cooling. Its pole (IM) plots at 110.6°E, 79.0°S ($A_{95}=8.4°$). The second direction is concordant with the IA group of Duff and Embleton (1976), which includes results from the unmetamorphosed Lakeview Dolerite dyke dated at 1116 Ma. The combined results from previous and present studies on the unmetamorphosed dykes give a pole (IAR) at 311.1°E, 9.5°N ($A_{95}=17.4°$). Poles IAR and IM suggest very high and middle palaeolatitudes at around 1100 and 1500 Ma, respectively. The new results agree with the most recent version of the Australian pole path by Idnurm and Giddings (1988), and do not challenge the concept of a single path for the continent.

1. Introduction

The nature of Precambrian geodynamics remains a matter of debate (Van der Voo and Meert, 1991). Questions such as "did plate tectonics play a role in dynamic processes of those remote times?" are still not answered unequivocally. Palaeomagnetism may ultimately help to settle such questions through independent studies on different continents, or on different cratons within one continent, by indicating whether parts of the crust have or have not moved relative to each other.

The Australian continent, like others, is a large block assembled from several cratons, usually separated by younger mobile belts or sedimentary basins. Accumulation of Precambrian palaeomagnetic poles for the assemblage commenced about 30 years ago. So far it has been possible to fit all poles to a single swath of apparent polar wander, as shown in successive reviews of Precambrian palaeomagnetism in Australia (McElhinny and Embleton, 1976; Embleton,

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1981; McWilliams, 1981; Idnurm and Giddings, 1988). Such a single apparent polar wander path (APWP) favours the idea that the present pattern of cratons has changed little, if at all, since its inception. But as noted by Idnurm and Giddings (1988), the number of Australian Precambrian poles is still small and their quality is generally not high (based on their criteria, only 4 poles out of 65 were nominated as key poles). Therefore the evidence for a single path is tentative and it is important to continue to refine the Australian APWP for the solution of geodynamic problems, as well as to improve it as a tool for dating.

In this paper we present the results of a new palaeomagnetic study on the mafic intrusions of the Mount Isa Inlier of northern Australia, with particular emphasis on unmetamorphosed dykes. The Inlier contains numerous intrusions of different ages on which only one, reconnaissance, palaeomagnetic study has been reported (Duff and Embleton, 1976). That study yielded, among other results, a 1116 Ma pole. The latter currently provides one of the few dated constraints on the 1400–900 Ma segment of the Australian APWP, although it is based on directions from only two dykes.

2. Geology and geochronology

The Proterozoic Mount Isa Inlier outcrops over an area of about 50,000 km^2 (Fig. 1). Its major Precambrian neighbours are the McArthur Basin to the west and the Arunta Block to the southwest. The former is part of the North Australian Craton (Plumb, 1979) which extends from the western Mount Isa Inlier to the Kimberley Block. The latter, Arunta Block, together with the Musgrave Block, is part of the system of Central Australian Mobile Belts (Plumb, 1979) which encloses the Yilgarn and Pilbara Blocks (the West Australian Shield), and forms the northern and eastern boundaries of the Gawler Block.

The Mount Isa Inlier is divisible into three main tectonic units (Carter et al., 1961; Blake, 1987): the Eastern and Western Fold Belts and the Kalkadoon–Leichhardt Belt (Fig. 1). The first two contain mostly Palaeo- and Mesoproterozoic cover sequences which comprise sedimentary and mafic rocks and granite intrusives; the third contains, in addition, metasediments and felsic igneous rocks of a Palaeoproterozoic basement. All sequences have been regionally metamorphosed during a period estimated as 1620–1500 Ma (Page et al., 1984). The metamorphic grade varies between different areas of the Inlier and is generally lower greenschist to upper amphibolite. The three tectonic subdivisions are intruded by numerous dykes and other mafic bodies, most of which are metamorphosed. Only two of the bodies have been dated: the Lunch Creek Gabbro by U–Pb zircon at 1740±25 Ma, and the Lakeview Dolerite dyke by Rb–Sr at 1116±12 Ma (Page, 1983). The latter is unmetamorphosed.

Ellis and Wyborn (1984) assigned the dolerite intrusions of each tectonic subdivision to broad age groups on the basis of deformation, metamorphism, geochronology and field relationships. The intrusions of the Western Fold Belt were assigned to three age groups (W1–W3), those of the Kalkadoon–Leichhardt Belt to two groups (B1, B2), and those of the Eastern Fold Belt to four groups (E1–E4). Increasing numbers denote decreasing age. Groups W3, B2 and E4 comprise unmetamorphosed dykes, and are considered to be probably equivalent (these dykes trend either north–south or east–west, and are distinguished in the field by well-rounded corestones).

3. Present work

We have extended Duff and Embleton's study (1976) to other mafic intrusives and across a larger area. A drawback of the present study was the lack of new radiometric ages; however, the classification of Ellis and Wyborn (1984) was expected to help assign at least a relative age to any primary remanence. Conversely, the remanence directions were expected to provide confirmation and refinement of the dyke groupings.

Over 60 localities were visited, representing most dyke groups of Ellis and Wyborn (1984).

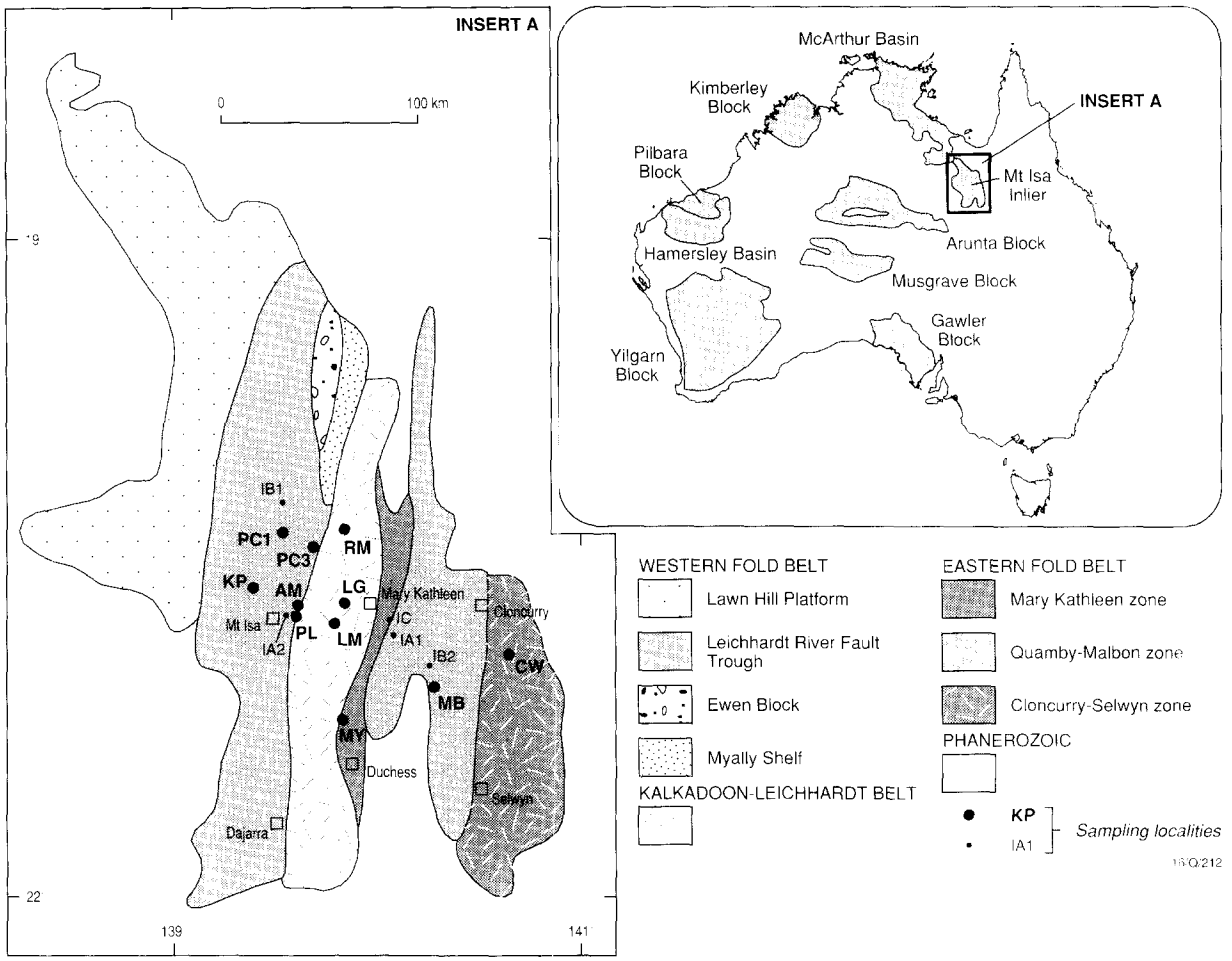


Fig. 1. Sampling localities shown on simplified tectonic map of the Mount Isa Inlier (after Blake, 1987). Dots show localities; acronyms refer to sites. The sites of two closely spaced dykes LM1 and LM2 are shown combined as site *LM* and KP1–5 sites are shown as *KP*. Locations of Lunch Creek Gabbro (1740 Ma) and Lakeview Dolerite (1116 Ma) and other dykes which were studied by Duff and Embleton (1976) are shown by smaller symbols. Map of Australia locates the Mount Isa Inlier in relation to other principal Precambrian units.

Many outcrops turned out to consist of boulders and isolated fragments, and the possibility of block rotation made these unsuitable for palaeomagnetic work. Nevertheless samples were collected from many such localities for rock magnetism. Also, many sampled sites were found to be magnetically unstable or to have inconsistent directions.

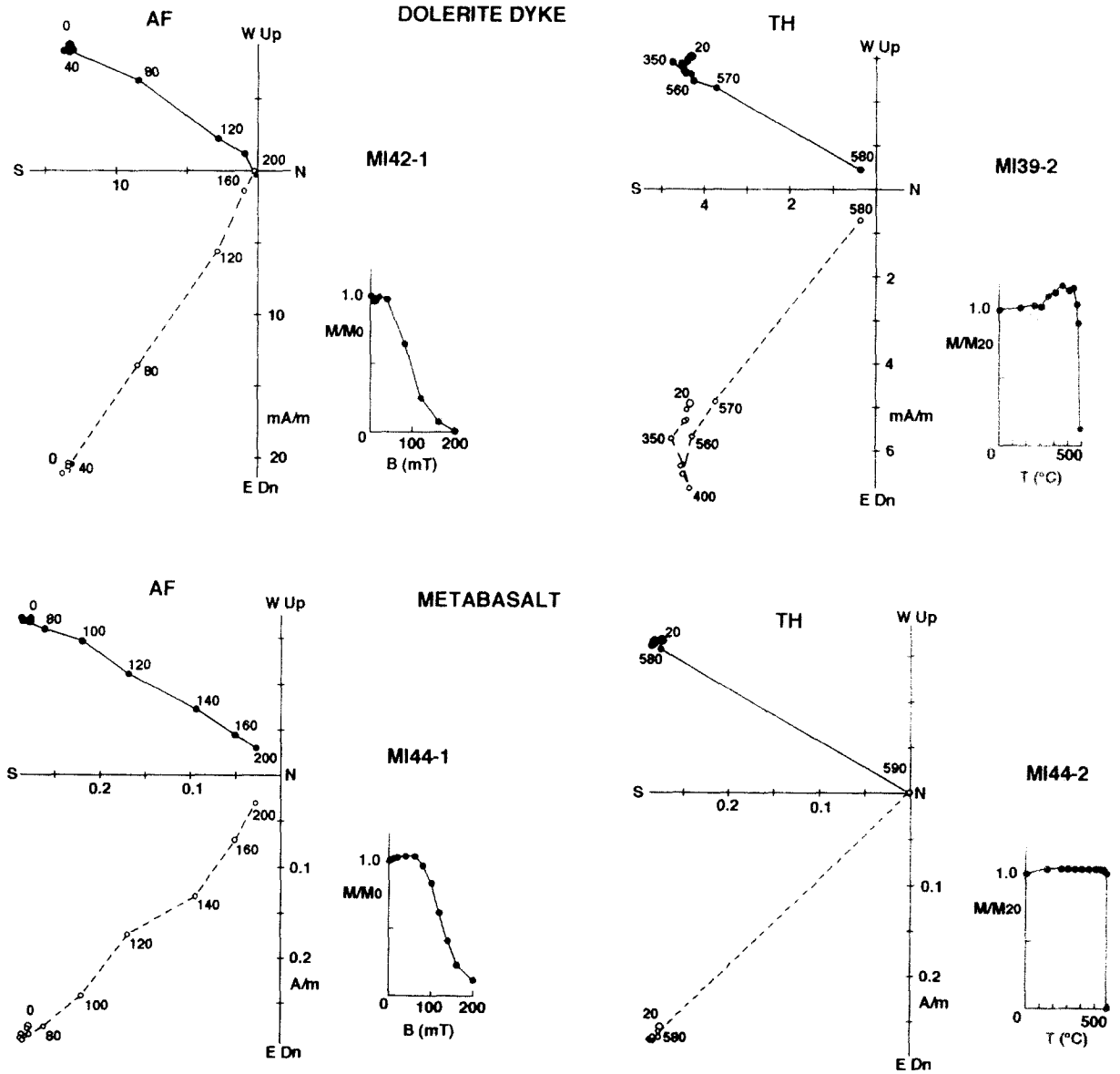
This paper reports the results from sixteen sites, representing eleven dykes and five plugs, where palaeomagnetic directions could be reliably determined. Samples were collected also from host rocks at six sites for baked contact tests: four were found magnetically stable. Fig. 1 shows the sampling sites on a tectonic map of the Inlier, and Table 1 lists their geographic co-ordinates. The sixteen results reported are fewer than we aimed for; nevertheless they increase threefold the number of mafic intrusions of the Inlier with measured palaeomagnetic directions.

4. Palaeomagnetic measurements

4.1. Experimental procedures

Six 2.5-cm diameter cores were drilled from most sites, the samples being oriented by means of sun and magnetic compasses. The samples were cleaned from surface contaminates by sand blasting and then cut to 2.2-cm height specimens.

Most remanence measurements were made at the Black Mountain Laboratory, Canberra, on either an ScT cryogenic magnetometer or a Digico spinner magnetometer; some were made at the Tokyo Institute of Technology on a 2G En-



(a) SITE PC1

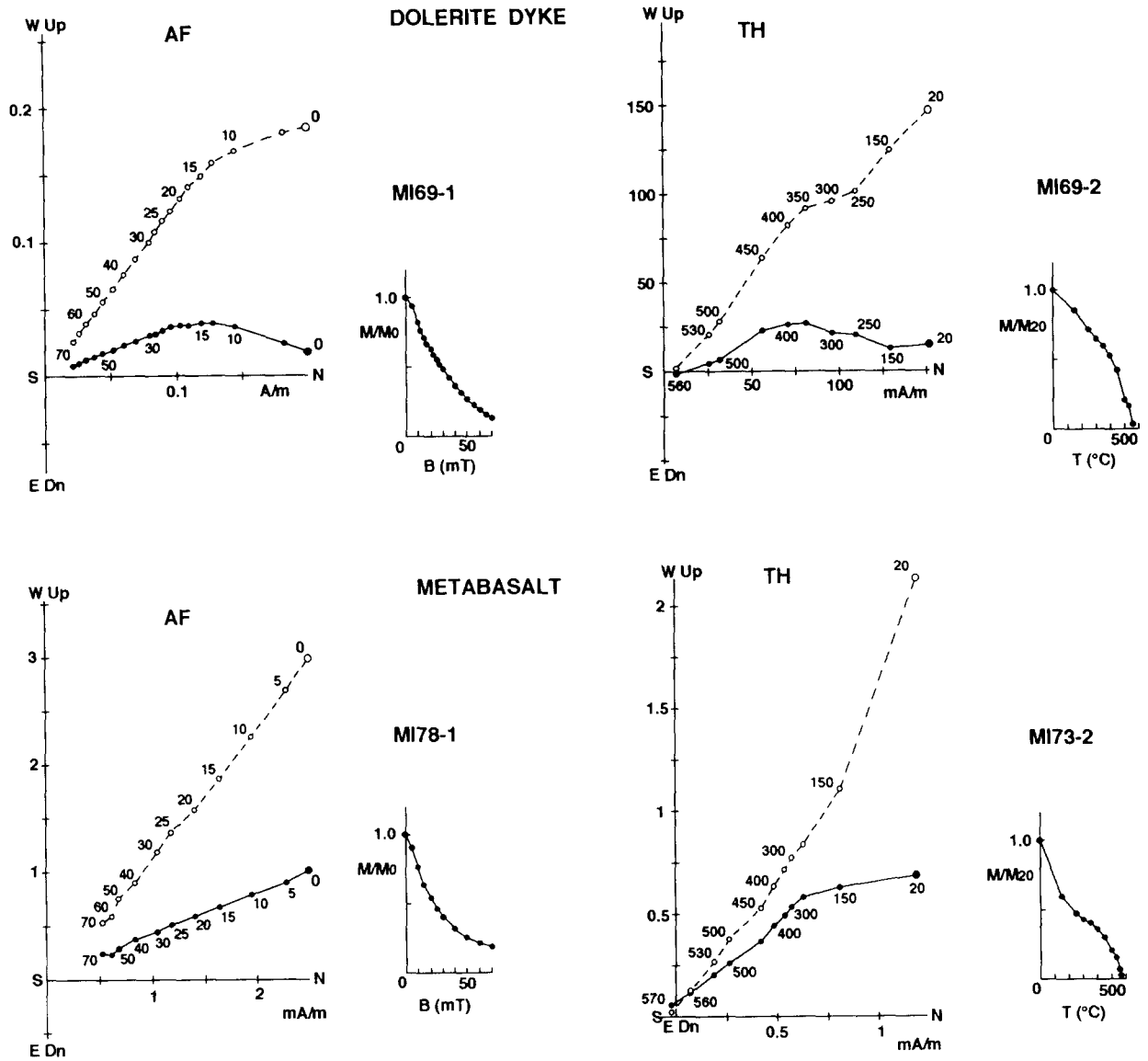


Fig. 2. Orthogonal projection plots and normalized intensity decay curves for AF/thermal demagnetizations: (a) site PC1 of W3 dyke and Cromwell Metabasalt; (b) site MB of E2 dyke and Cone Creek Metabasalt. Filled/open symbols represent horizontal/vertical projections.

terprises cryogenic magnetometer 740-R or a Schonstedt spinner magnetometer SSM-1A. At least one specimen from each sample was demagnetized progressively in alternating fields (AF) of usually up to 100 mT peak intensity, using a Schonstedt or a home-made tumbler demagnetizer. In addition, one specimen per site was usually demagnetized thermally in a large furnace described by McElhinny et al. (1971), in a Schonstedt demagnetizer or in a home-made thermal demagnetizer. Exceptions were for samples from net-veined complexes between doler-

Table 1
Site localities of mafic dykes, plugs, and host rocks from the Mount Isa Inlier

Site	Rock type	Long. (E)	Lat. (S)	Map (1:100,000)	Grid
KP1	dolerite plug (W2)	139°23′	20°36′	Mount Isa	322221
KP2	dolerite plug (W2)	139°23′	20°35′	Mount Isa	312226
KP3	dolerite plug (W2), Sybella Granite	139°22′	20°35′	Mount Isa	304226
KP4	dolerite plug (W2), Sybella Granite	139°22′	20°35′	Mount Isa	302228
KP5	dolerite plug (W2)	139°22′	20°35′	Mount Isa	293234
LG	young dolerite dyke (B2)	139°50′	20°41′	Mary Kathleen	791121
LM1	young dolerite dyke	139°48′	20°46′	Mary Kathleen	743019
LM2	young dolerite dyke	139°48′	20°46′	Mary Kathleen	743019
PL	young dolerite dyke	139°36′	20°43′	Mary Kathleen	538079
AM	young dolerite dyke	139°37′	20°40′	Mary Kathleen	553148
PC1	dolerite dyke (W3), Cromwell Metabasalt	139°32′	20°21′	Prospector	468501, 478506
PC3	young dolerite dyke	139°40′	20°25′	Prospector	616428, 615425
RM	dolerite dyke	139°50′	20°20′	Prospector	776529
MY	young dolerite dyke at Myubee Igneous Complex	139°50′	21°12′	Duchess	782546
MB	dolerite dyke (E2), Cone Creek Metabasalt	140°17′	21°03′	Kuridala	258730
CW	dolerite dyke	140°38′	20°55′	Cloncurry	623874

ite plugs and the Sybella Granite, for which thermal demagnetization was mostly used, and for some samples with discordant directions. The latter samples were first partially demagnetized by AF and then demagnetized thermally to check if lightning-induced components were present (Cox, 1961; Graham, 1961). This procedure did not reveal any new components. The number of demagnetization steps was between ten and twenty.

The characteristic remanence components were usually determined from linear segments using the principal component analysis technique (Kirschvink, 1980). The acceptance criterion for linear elements was a maximum angular deviation (MAD) of 5° or less. The results of some samples at two sites were analyzed by great-circle fitting; the MAD values were $< 7°$.

4.2. Metamorphosed dolerite dykes

Many metamorphosed dykes had well-defined remanences that comprised one or two components in addition to a very soft component. Representative orthogonal projection plots for AF and thermal demagnetizations are shown in Fig. 2 for two sites where both the dyke and its metabasalt host were sampled.

Fig. 2a shows the results from a W3 dyke and the Cromwell Metabasalt (a member of the widespread Eastern Creek Volcanics) at site PC1. The latter was sampled about 200 m from the dyke contact. The dyke and metabasalt gave single component remanences with closely similar directions. The magnetizations were very stable, as shown by high median destructive fields of 80–120 mT and unblocking temperatures (Tb) just below 580°C. Although this dyke has been classified as unmetamorphosed (W3), the remanence directions suggest that it is overprinted.

Fig. 2b shows the results for an E2 dyke and its host, the Cone Creek Metabasalt, at site MB. The latter was sampled at six spot locations ranging from 5 to 100 m of the dyke margin. The dyke and metabasalt again gave, besides very soft components, single component remanences with

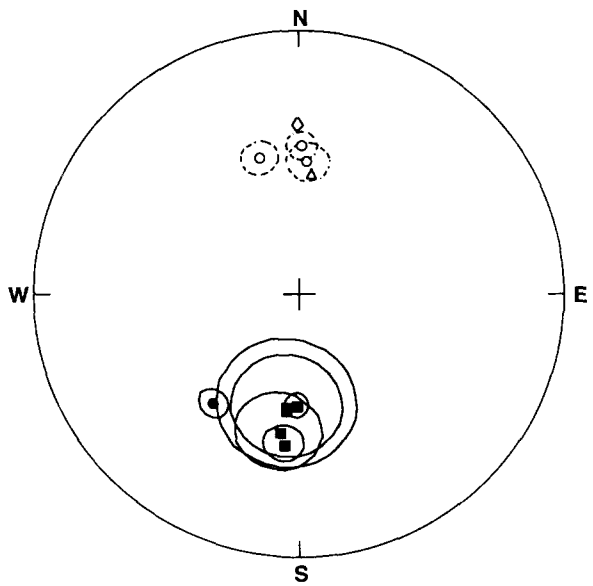


Fig. 3. Equal area plots of site mean remanence directions for metamorphosed dolerite plugs (square symbols), dykes and metabasalt host rocks (circle symbols). Open/filled symbols denote upper/lower hemisphere. Diamond indicates axial dipole field direction and triangle indicates present field direction. Circles show 95% confidence.

similar directions. At this site the coercivities are relatively low and the unblocking temperatures have a distributed spectrum.

Two other metamorphosed dykes (sites RM and CW) gave remanence directions similar to those of the E2 dyke and Cone Creek Metabasalt. Although all these directions are similar to the present axial dipole direction, we consider a viscous remanent magnetization of recent origin unlikely because the remanences were stable up to fairly high AF and thermal demagnetization steps. The samples showed no evidence of recent chemical weathering, and there are no reasons to suspect any other kind of recent overprinting. Equal area plots of the site mean palaeomagnetic directions are shown in Fig. 3 along with those for the metamorphosed dolerite plugs at Kitty Plain, which are discussed in Sect. 4.4. The site mean remanence data are summarized in Table 2 under Group 1.

4.3. Unmetamorphosed dolerite dykes

Thirteen unmetamorphosed dykes were examined in the field. Most were found to consist of boulders and fragmented outcrops with various probabilities of block rotation. Twelve sites were sampled. The remanences of all dykes were stable and had a single major component accompanied usually by minor soft components, but we report here results from only seven north–south-trending dykes with within-site directional consistencies that preclude block rotation.

Orthogonal projection plots of AF and thermal demagnetization are shown for two of the dykes (sites PL and AM) in Fig. 4. Their remanence directions are characteristically near-vertical and downward. Equal-area plots of the seven site mean directions are shown in Fig. 5; the results are summarized in Table 2 under Group 2.

The host rocks of several unmetamorphosed dykes were sampled for baked contact tests. Most proved unsuitable because of unstable remanences or highly scattered directions; none gave a fully satisfactory result. The best evidence for remagnetization by baking was obtained from site LG. At that site a red sandstone sampled 1 m from the dyke contact gave a direction that appears to be intermediate between (1) the steeply inclined, downward directions of the dyke, and (2) Group 1 type directions in the sandstone beyond 3 m of the contact. Lack of outcrop prevented detailed sampling within 1 m of contact.

4.4. Metamorphosed dolerite plugs

Kitty Plain, 20 km northwest of Mount Isa, contains several metamorphosed dolerite plugs, which form low, boulder-strewn hills (Ellis and Wyborn, 1984). The area is part of the Sybella Batholith, which has U–Pb zircon dates of 1671 Ma (Page, 1981), 1660 Ma and 1665 Ma (Connors and Page, 1992) for its different plutons. The plugs and granite are probably coeval (Ellis and Wyborn, 1984): some plugs form net-veined complexes with the granite (Blake, 1981).

Five sites were sampled, including two that contain net-veined complexes. At three sites the remanences were stable and straightforward to

Table 2
Site mean paleomagnetic directions and paleomagnetic poles

Site	AF	TH	N	Dec	Inc	α_{95}	P_{lat}	P_{lon}	dp	dm
Group 1 (pole IM)										
KP1	4/4		4	188.6	45.9	12.8	−79.7	91.3	10.4	16.4
KP2	5/5		5	186.4	54.2	16.6	−74.8	118.8	16.3	23.3
KP3 [a]	7/10	11/14	18	181.4	55.2	3.9	−74.8	135.0	4.0	5.6
KP4 [a]		7/8	7	186.0	42.3	6.0	−83.2	85.4	4.5	7.4
KP5	2/3	1/1	3	186.6	54.9	20.9	−74.1	119.5	21.0	29.6
PC1	6/7	2/2	8	218.2	46.0	4.5	−54.6	68.3	3.7	5.8
RM	6/7	1/1	7	4.0	−48.1	6.4	80.4	298.1	5.5	8.4
MB	7/10	2/2	9	343.7	−45.0	5.5	74.1	26.5	4.4	6.9
CW	5/5	4/4	9	1.3	−42.8	4.7	85.9	304.4	3.6	5.9
Mean [b]: Dec = 6.1, Inc = −49.0, α_{95} = 6.9, P_{lat} = −79.0, P_{lon} = 110.6, A_{95} = 8.4										
Group 2 (pole IAR)										
LG	4/5	1/1	5	285.2	64.9	7.5	−5.2	98.4	9.7	12.0
LM1	3/4	1/1	4	267.4	79.2	11.4	−20.3	117.5	20.7	21.7
LM2	3/3	1/1	4	283.2	74.7	10.4	−12.0	111.2	17.2	18.9
PL	6/6	1/1	7	76.7	82.1	3.5	−16.4	155.4	6.7	6.8
AM	5/5	1/1	6	42.2	66.8	5.6	10.5	166.0	7.7	9.3
PC3	4/4	1/1	5	54.7	71.4	4.4	0.8	166.8	6.8	7.8
MY	6/6	1/1	7	293.2	78.4	6.6	−11.3	119.0	11.7	12.4
Mean [b]: Dec = 333.1, Inc = 82.8, α_{95} = 12.8, P_{lat} = 8.6, P_{lon} = 313.4, A_{95} = 23.7										
Mean [b]: including IA directions of Duff and Embleton (1976): Dec = 322.8, Inc = 82.5, α_{95} = 9.6, P_{lat} = 9.5, P_{lon} = 311.1, A_{95} = 17.4										
Poles of Duff and Embleton (1976) [c]										
IA Group from Lakeview Dolerite: N = 3, Dec = 297, Inc = 80, P_{lat} = −12, P_{lon} = 124, A_{95} = 11										
IB Group from Gabbroic dykes in the Western and Eastern Fold Belts: N = 2, Dec = 208, Inc = 60, P_{lat} = −53, P_{lon} = 102, A_{95} = 11										
IC Group from Lunch Creek Gabbro: N = 6, Dec = 152, Inc = 50, P_{lat} = −63, P_{lon} = 201, A_{95} = 9										

$n1/n2$ under AF and TH is number of specimens used in calculation/subjected to AF and thermal demagnetization, respectively. N is total number of specimens used for site statistics. Dec and Inc are declination and inclination. α_{95} and A_{95} are 95% confidence circles for site direction and group mean paleomagnetic pole, respectively. P_{lat} and P_{lon} are latitude and longitude of south pole position. dp and dm are radii of 95% confidence ellipse of VGP.

[a] Calculation was made by converging great-circle method.

[b] Choice of pole from the pole–antipole pair was made in conformity with the Australian APWP of Idnurm and Giddings (1988).

[c] Reproduced here for easy comparison. Note that in Duff and Embleton (1976) IB and IC were taken as north poles. N is number of sites for the mean.

interpret. The site mean directions are similar to the directions from metamorphosed dykes (Table 2 and Fig. 3).

The demagnetization behaviours of two of the sites associated with net-veined complexes, KP3 and KP4, were more complicated. Eighteen dolerite samples, including two dolerite xenoliths at 5 m from the dolerite contact, and six granite samples were collected from site KP3. The latter samples were 10 cm to 50 m from the contact

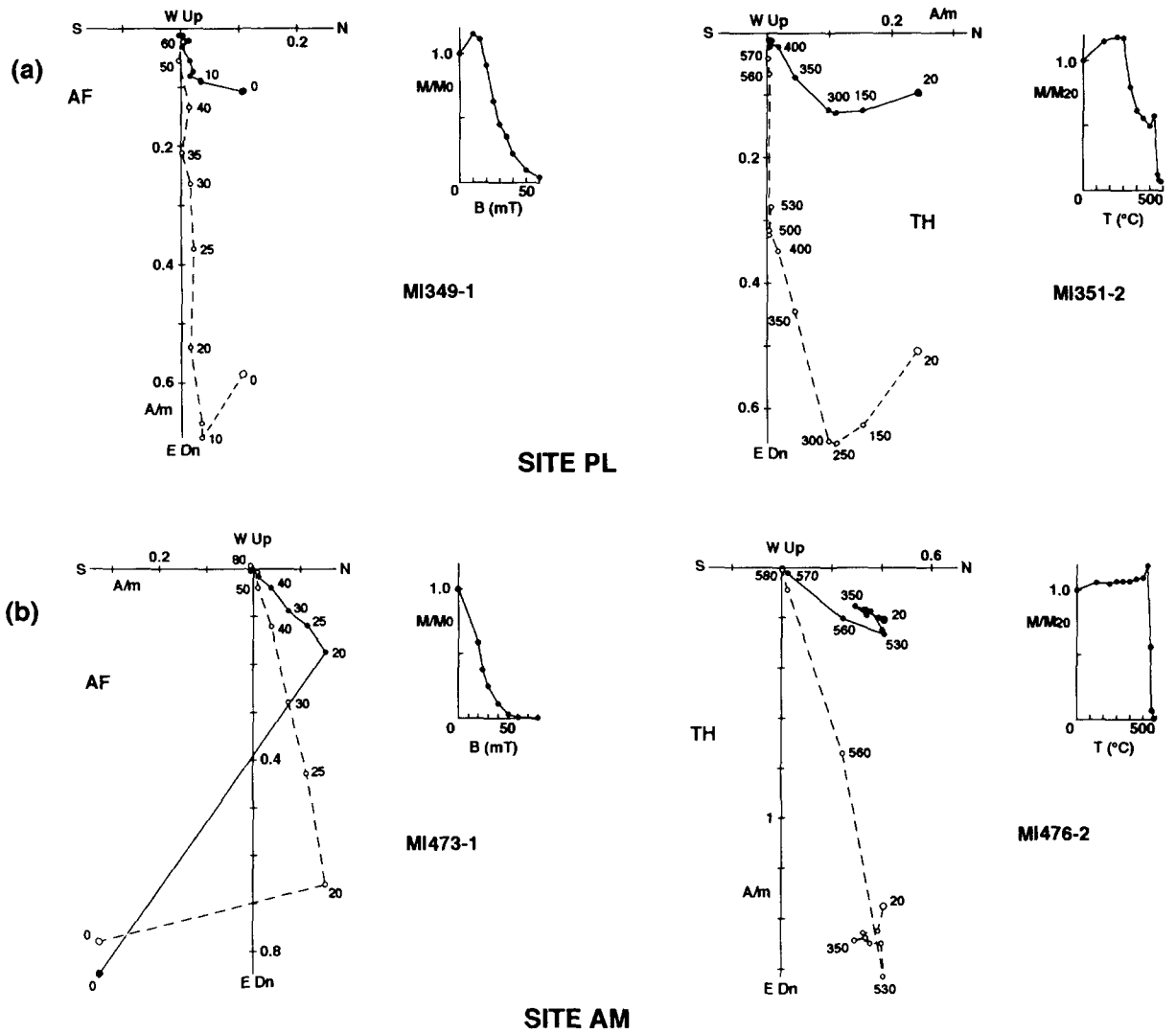


Fig. 4. Orthogonal projection plots and AF/thermal demagnetization curves for unmetamorphosed young dykes at (a) site PL and (b) site AM. Symbols are as in Fig. 2.

zone. Fig. 6 shows some of the results.

In Fig. 6a thermal demagnetization results are plotted for a dolerite sample (MI266-1) from the edge of the contact zone. The orthogonal projection diagram contains two linear segments: the first gives a northerly, upward directed component with Tb less than 500°C, the second a southerly, downward directed component with Tb between 500°C and 580°C. This is the only sample from site KP3 for which both components were resolved: for example, Fig. 6b shows alternating field (AF) demagnetization results from one of the dolerite xenoliths (MI33-1) in which the remanence contains also two components, but where only the higher coercivity component is clearly resolved.

The component with upward inclination was isolated as a linear segment in the orthogonal plots of seven dolerite and three granite samples. It was always accompanied by the main southerly downward directed component except in one granite sample (MI276-1) from near the doler-

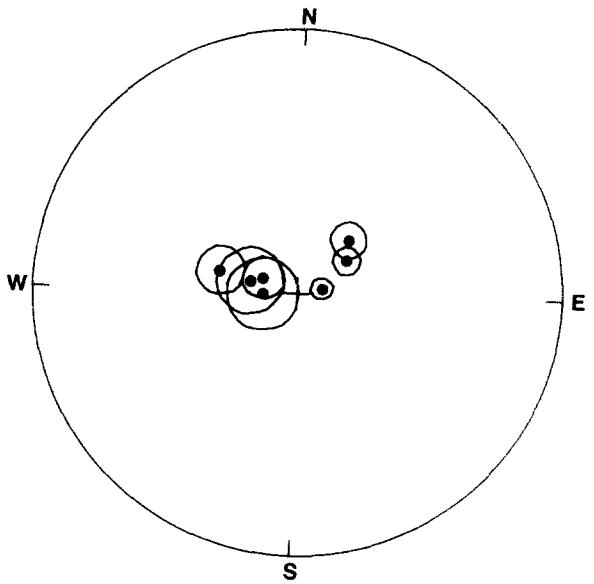


Fig. 5. Equal area plots of site mean remanence directions for unmetamorphosed dolerite dykes. Notations are as in Fig. 3.

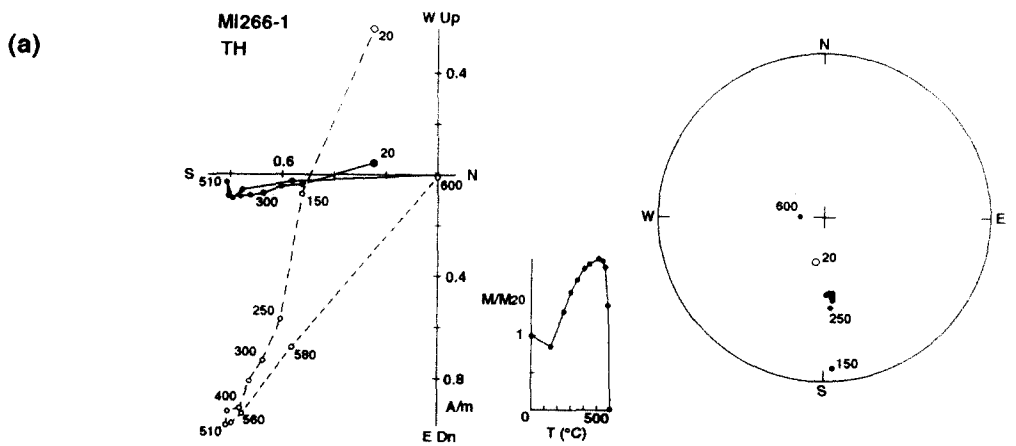
ite contact that contained only the upward directed component (Fig. 6c).

In contrast, Figs. 6d–6f illustrate AF and thermal demagnetization results from dolerite plug samples for which orthogonal projection diagrams contain no linear segments, except perhaps for the low-intensity demagnetization steps. The sample MI35-1 in Fig. 6d was collected 50 m from the contact; other samples (MI267-1, MI269-1) were obtained close to the contact. In each case a great circle provides an excellent fit to directions on the equal-area projection. The circle intersections give directions similar to those of high Tb and Hc components in Figs. 6a and 6b, and to those collected from the other plugs.

Besides the two end members of the demagnetization great circles, characterized by moderate upward and downward inclination, there is some evidence for a possible third component of steep downward inclination at site KP3. Fig. 6g shows this direction in a dolerite sample (MI278-2) about 50 cm from the granite sample of Fig. 6c. (The dolerite and granite samples were 10 cm and 50 cm, respectively, from the contact zone.) This direction was found as a linear segment on the orthogonal diagram in only two samples.

Fig. 7 shows the three directional groups at site KP3. The first, shown in Fig. 7a, contains seven directions from orthogonal plot analyses and eleven from great-circle intersections. Also shown is their mean, calculated using the method of McFadden and McElhinny (1988). This component was found only in dolerite samples; its absence in granite samples may be due to a lower resistance of the granite to subsequent overprinting. Where both the upward and downward directed components were present, the remanence associated with the downward component was always the more stable of the two in both AF and thermal demagnetizations.

Fig. 7b shows the upward inclined directions.



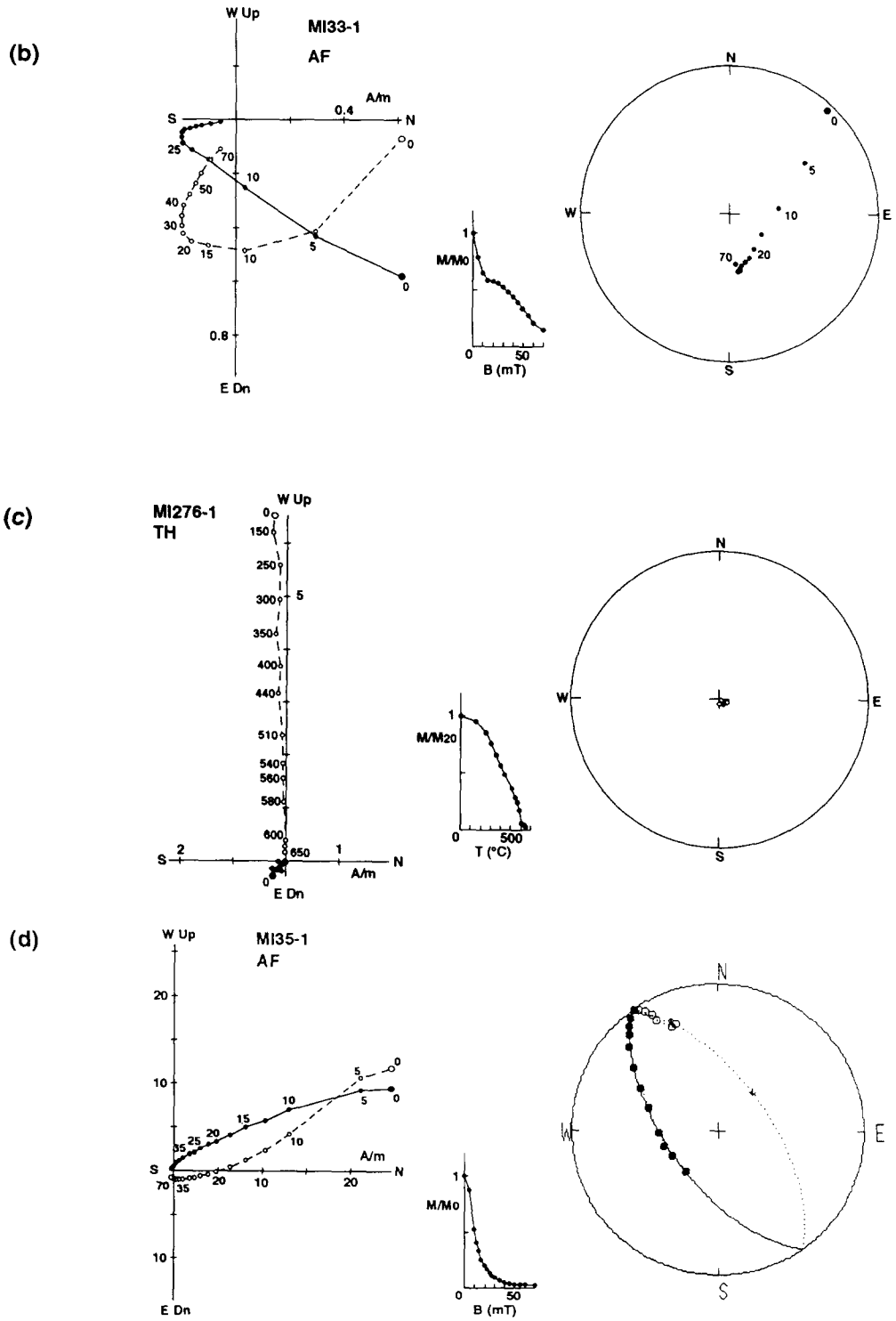


Fig. 6 (legend on p. 252)

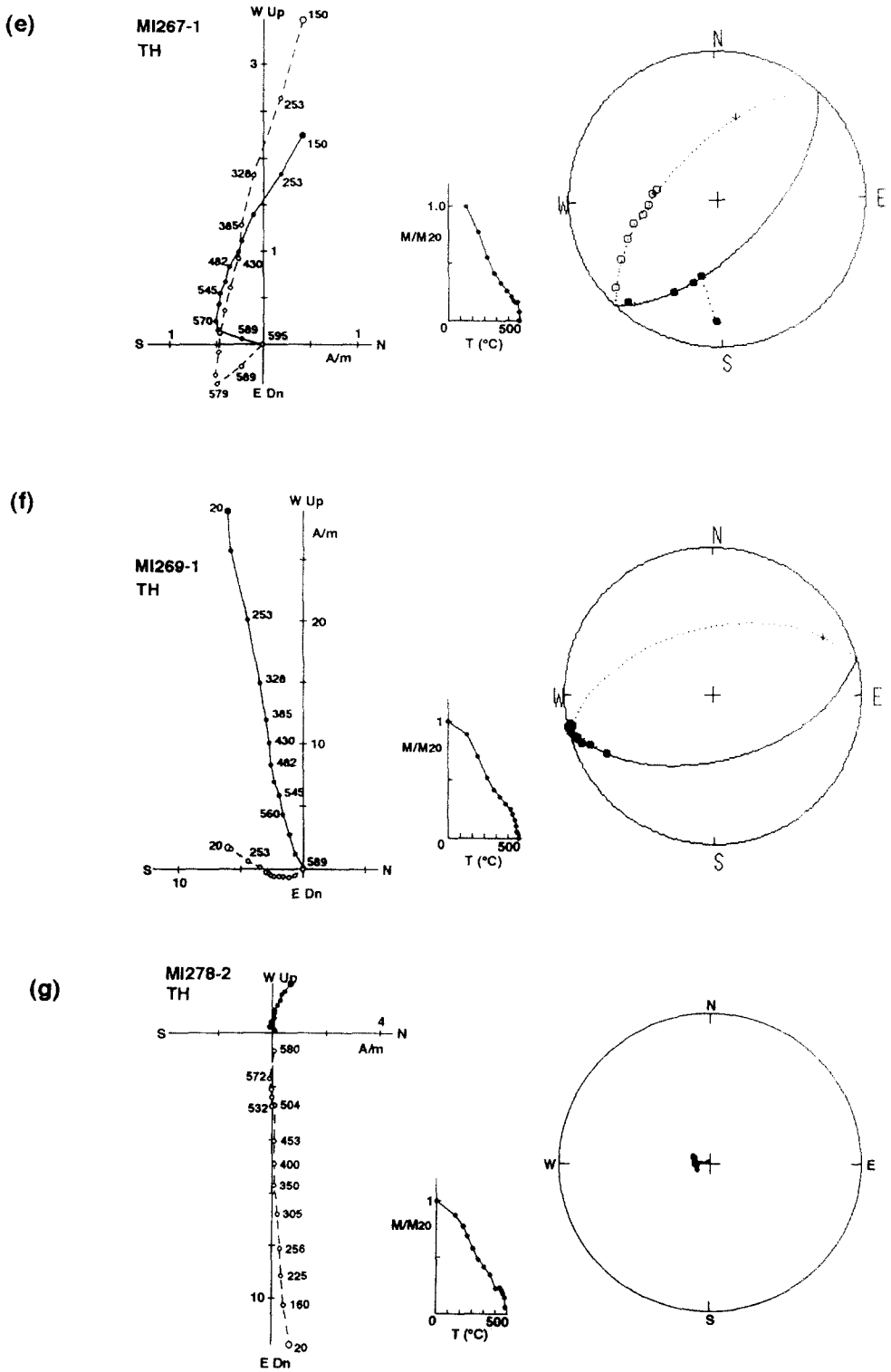


Fig. 6. Various demagnetization data from Kitty Plain site KP3: (a) and (b) are for dolerite samples in which the downward directions were defined by linear segments on orthogonal plots; (c) a granite sample that has a single vertical upward component; (d), (e) and (f) are for dolerite samples in which no linear segments were recovered but which gave excellent fits to great circles; (g) a dolerite sample which shows a vertical downward direction. Symbols are as in Figs. 2 and 3.

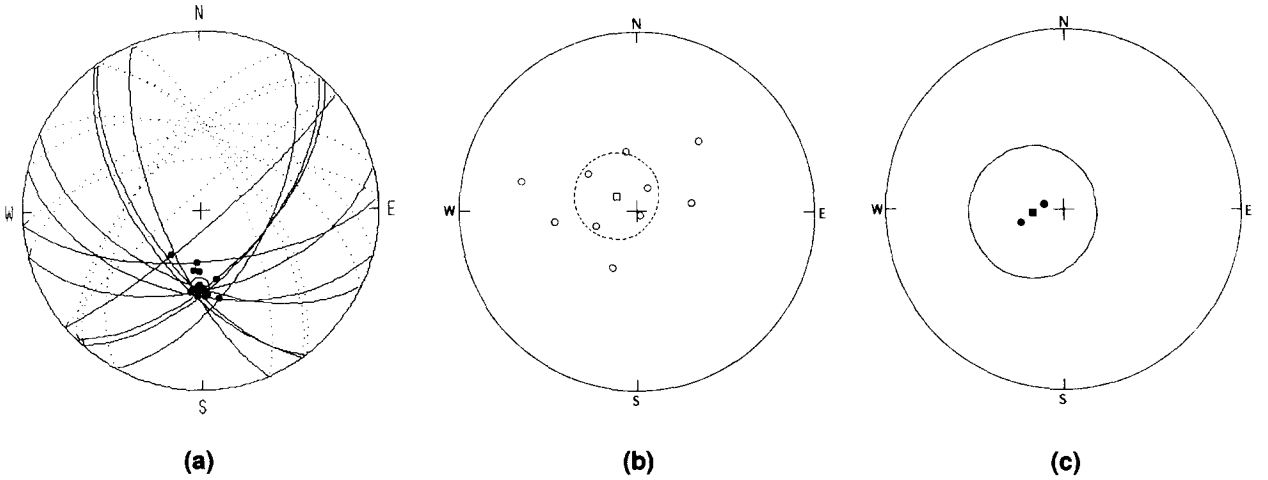


Fig. 7. Directional groups from Kitty Plain site KP3 obtained by principal-component and great-circle analyses: (a) well defined downward mean direction obtained from seven linear and eleven great-circle results from dolerite samples; (b) upward directions obtained from dolerite and granite samples; (c) almost vertical downward directions obtained from two linear results. Symbols as in Fig. 3. Circles show 95% confidence.

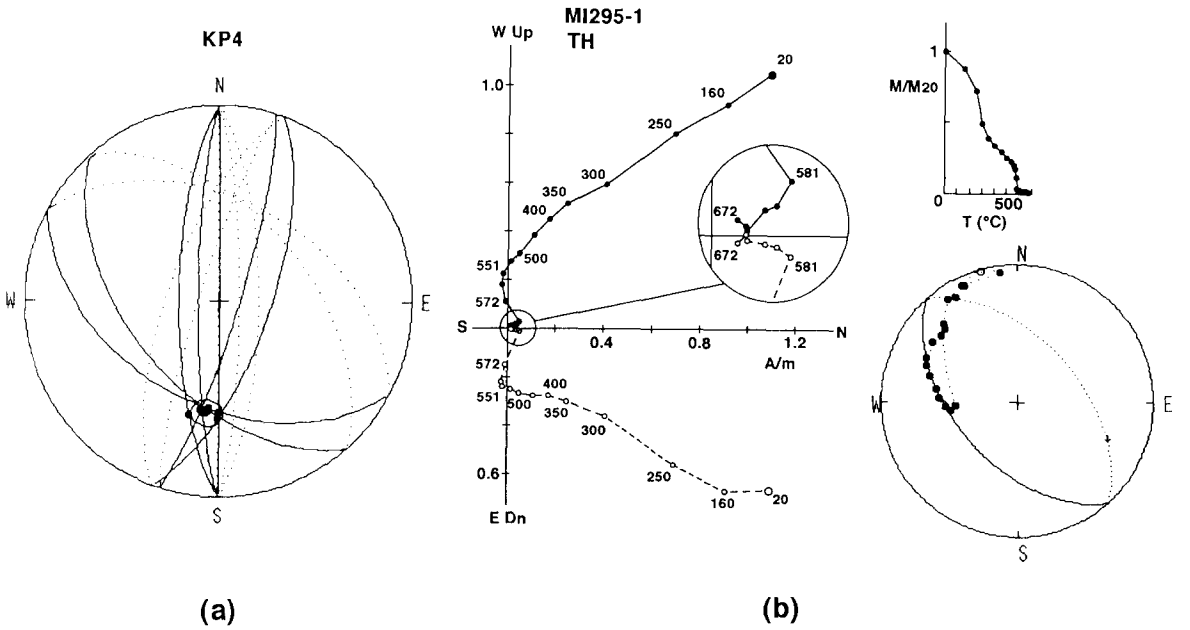


Fig. 8. (a) Directional group from Kitty Plain site KP4 obtained from great-circle intersections, and (b) an example of a thermal demagnetization orthogonal plot for a granite sample in which the low and high Tb components are almost parallel. Symbols are as in Figs. 2 and 3.

The group is dispersed, and it is this dispersion that allows the great-circle method to be used for isolating the harder component. The third group, comprising two steeply inclined downward directions, is shown in Fig. 7c. Only one sample remained difficult to allocate to the three groups. The AF demagnetization path of this sample was an equatorial great circle (remanence direction was downward and the trend was from west to east). The remanence of this sample may be related to the vertical downward directions because the great circle passes near the directions shown in Fig. 7c.

Great circle analysis of the Kitty Plain contact site KP4 is shown in Fig. 8a. Four cores were drilled from both dolerite and granite, all within one metre of the contact. Progressive thermal demagnetization was used on all samples. The orthogonal projection plots contained no linear segments so that set point constraints are not available, but the great circle intersections can be clearly interpreted in terms of a southerly downward dipping component (Fig. 8a). An interesting exception in one of the granite samples is shown in Fig. 8b where thermal demagnetization reveals low and high Tb components that are almost parallel. The high Tb component survives above 650°C, suggesting hematite as the carrier. Such a feature may not be uncommon in contact zones as has been described by Schwarz and Buchan (1989) at Sudbury, Canada, and by Schmidt (1990) at the Sydney Basin, Australia.

5. Discussion

The present study yields two main groups of remanence directions. The first was obtained from metamorphosed intrusions, including the dolerite plugs of Kitty Plain, and gives the pole IM at 110.6°E, 79.0°S (95% confidence angle $A_{95}=8.4°$). Since the intrusions are from the different age groups of Ellis and Wyborn (1984), IM clearly represents a metamorphic overprint. The youngest recognized period of regional metamorphism in the Mount Isa Inlier was at 1620–1500 Ma (Blake, 1987), with the likely date of 1554 Ma for the last major metamorphic peak (Connors and Page, 1992). This peak reached locally sillimanite facies to partial melting and was followed by another, smaller, peak (Connors and Page, 1992). Thus the age of IM lies most probably between 1554 Ma and the end of metamorphism, ~1500 Ma. This age is consistent with the moderate amount of scatter in the remanence directions, which suggests that the magnetization post-dates the last phase of major deformation of the Inlier—the latter is estimated to have ended at 1550 Ma (Blake, 1987). We note that K–Ar ages of 1500–1400 Ma have been obtained from widespread granites in the Mount Isa Inlier (Richards et al., 1963; Page, 1978). These have been identified as cooling ages, suggesting the possibility that the remanence post-dates the metamorphism.

The pole IM was plotted on the Australian APWP in Fig. 9. The path is after Idnurm and Giddings (1988), updated by a new pole KG (Camacho et al., 1991) and by an amendment to the pole GC. The original calculation of GC (Facer, 1971) appears to contain an error and its recalculated position of 18°S, 127°E is close to IAR. The pole IM plots on the path segment that contains the Gawler Block poles GR from the Gawler Range Volcanics (Chamalaun and Dempsey, 1978) and GA from basaltic dykes (Giddings and Embleton, 1976). The Volcanics have been dated by U–Pb zircon at 1592 ± 2 Ma (Fanning et al., 1988), in apparent disagreement with the age of IM. However, a recent fold test indicates that the pole GR is not primary as previously believed (P.W. Schmidt, private commun., 1992). The last major orogeny in the Gawler Block, the Kimban Orogeny, ended before the extrusion of the Gawler Range Volcanics, but according to K–Ar dates which are generally 1550–1450 Ma and in some parts 1400 Ma (Webb et al., 1986), high temperatures were maintained until much later. Also the Rb–Sr ages from different members of the Gawler Range Volcanics cluster around 1525 Ma. Deformation of the block (the Wartakan Event at ~1500 Ma) occurred during this period, causing local warping of the Gawler Range Volcanics (Fanning et

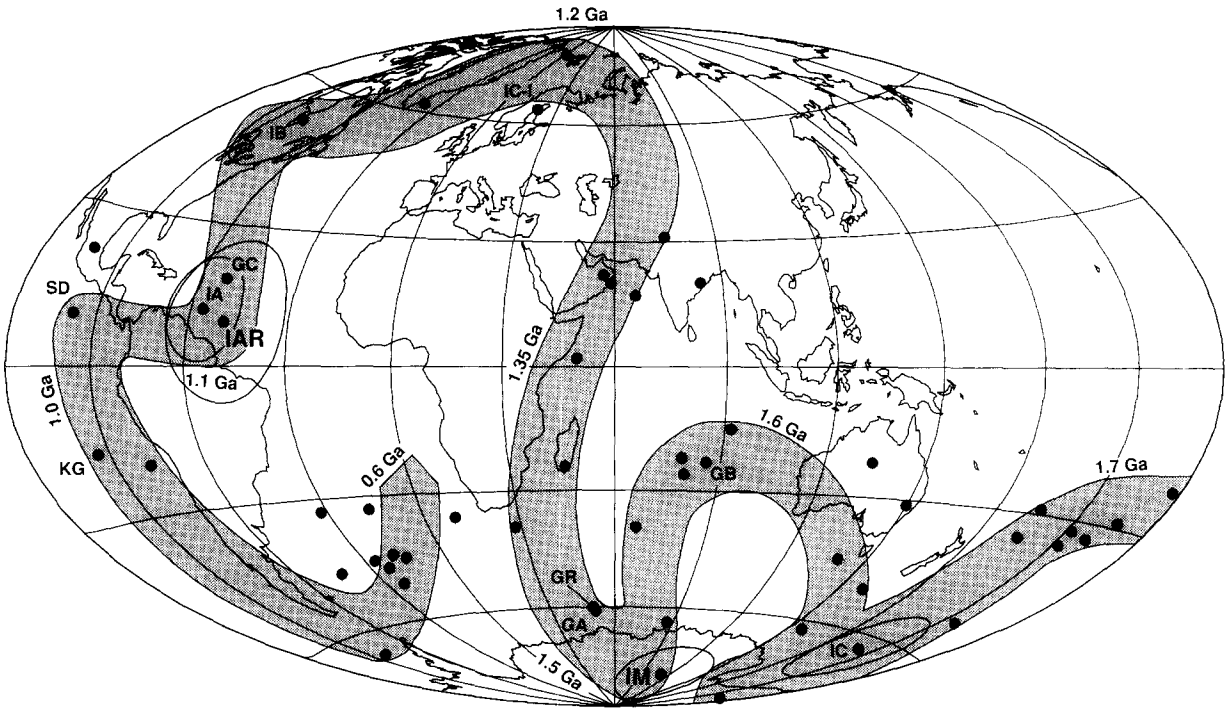


Fig. 9. Palaeomagnetic poles *IM*, from metamorphosed dykes, metabasalts, and Kitty Plain dolerite plugs (Group 1), and *IAR*, from unmetamorphosed dykes (Group 2), plotted on the Australian APWP. The path is after Idnurm and Giddings (1988), updated by new pole *KG* (Camacho et al., 1991) and by an amendment to pole *GC* from the 1100–1200 Ma Giles Complex. Duff and Embleton's (1976) poles *IA*, *IB*, and *IC* from the Inlier are shown for comparison.

al., 1988). If the remanences post-date this deformation, the Gawler Block poles GR and GA would be consistent with IM.

It is not clear, however, how the overprint remanences in the Gawler Block were acquired, given that the K–Ar closure temperatures for biotite are 200–350°C (Layer et al., 1989), whereas the unblocking temperature spectra for GR have their major peaks at or near the Curie temperature of magnetite (Chamalaun and Dempsey, 1978). Theoretical models for single-domain thermoviscous remanence would predict much lower unblocking temperatures (e.g., Pullaiah et al., 1975), and experimental evidence on multidomains appears to suggest that only the spectral tails would reach the Curie temperature (Worm et al., 1988). Possible explanations are that the deformation event was associated with heating or that experiments at the laboratory time scale on multidomain magnetite do not indicate the high-temperature characteristics of thermoviscosity acquired over longer, geological times.

The position on the path of the previously determined poles IC from the Lunch Creek Gabbro is problematic. Two independent studies on the gabbro gave consistent remanence directions (Duff and Embleton, 1976; N. Roberts, quoted in Piper, 1988). However, IC lies far from the overprint pole IM. It is unlikely that the primary remanence in the gabbro survived the last regional metamorphism (the pole also falls far from the 1740 Ma path segment appropriate for the U–Pb zircon age of the gabbro), suggesting later remagnetisation. The Lunch Creek Gabbro is adjacent to rocks in which the Rb–Sr systems were disturbed after 1500 Ma (Page, 1983). Samples from one of those, a skarn, gave well defined isochrons with dates of ~1300 and ~1200 Ma, although a single sample from the gabbro itself gave K–Ar and Rb–Sr ages of ~1440 Ma (Page, 1983). If the remanence of the Lunch

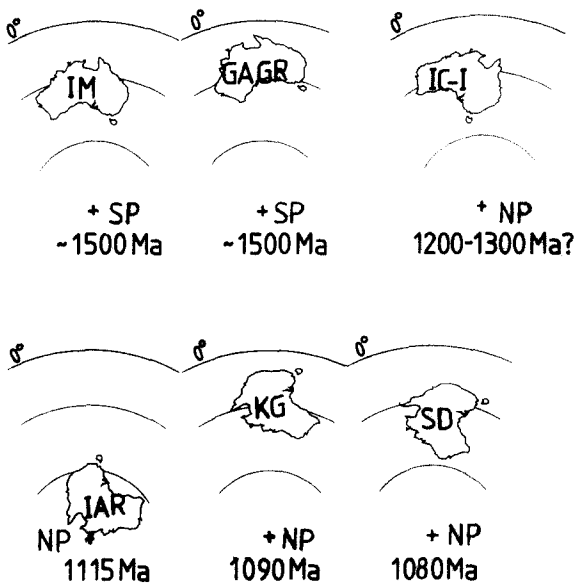


Fig. 10. Palaeolatitudes of Australia as indicated by poles from the Mount Isa Inlier, and the Arunta, Gawler and Musgrave Blocks. In each figure, a geographic pole is shown by a cross, and *NP* and *SP* indicate that the pole is interpreted as north pole and south pole, respectively.

Creek Gabbro was acquired after the last regional metamorphism, then the antipole of IC would be more appropriate for the pole path (IC-I, Fig. 9).

The second directional group was obtained exclusively from unmetamorphosed dykes. The grouping supports the belief that these dykes were intruded during one general period of magmatic activity (D.H. Blake, pers. commun., 1992). The dispersion of directions may be due to tectonic disturbances (some dykes are offset by or end at faults) or to small age differences between the dykes (e.g., recording secular variations or polar wander).

The remanence directions of the second group are similar to those obtained by Duff and Embleton (1976) for the unmetamorphosed Lakeview Dolerite and a second, undated, dyke. The results from the six dykes of the present study were combined with Duff and Embleton's results (IA group) to provide the updated pole 311.1°E, 9.5°N ($A_{95}=17.4°$) for unmetamorphosed dykes (the antipodal pole 131.1°E, 9.5°S was not adopted because of incompatibility with the established APWP). Since the results for the two IA dykes were not available separately, we used in the calculation a weighting factor of two for the IA direction and one for each present dyke. IAR is one of the few constraining poles for the 1400–900 Ma segment of the Australian APWP. Other poles with similar ages are KG from the Kulgera dyke swarm of the Musgrave Block, and a preliminary pole SD from the Stewart dyke swarm of the Arunta Block (Fig. 9). IAR is older than the other poles, in agreement with the path. However, SD and KG which have similar Sm–Nd ages, 1076±33 Ma and 1090±32 Ma (Zhao and McCulloch, 1994), are separated by about 30°. Central Australia was affected during the Carboniferous by the Alice Spring Orogeny, which produced folds, uplift and thrust-nappes. The remanences over a large region, including the Kulgera dykes (Camacho et al., 1991), were partly or completely overprinted. The orogeny left a strong structural imprint in the Stuart dykes area and the ~200 Ma discrepancy between the Rb–Sr (897±9 Ma) and Sm–Nd ages of the dykes could be due to partial resetting of the Rb–Sr system. Because of the complex deformation and overprinting, further conclusions would need to await the completion of the palaeomagnetic study on these dykes.

Kitty Plain site KP3 yielded, in addition, two less well established remanence directions, both with steep inclination. The first is directed upward and is highly scattered. Its origin is not clear, but a lower unblocking temperature than that of the component identified as the metamorphic overprint suggests that the remanence is younger than the metamorphism. The second remanence resembles the Group 2 directions of the unmetamorphosed dykes in its steep downward inclination. It is based, however, on linear results from only two samples and its existence is therefore not well established. The direction may represent, for example, a pseudocomponent produced by two remanences with matching or similar demagnetization characteristics.

Several samples were collected from unmetamorphosed dykes for geochronological dating. All showed some degree of alteration in thin sections, and only the two freshest samples were

submitted for K–Ar dating. The first, from site LG, gave the ages 868 ± 12 Ma and 1135 ± 62 Ma, respectively for plagioclase and pyroxene; the second, from site MY, gave the age 1179 ± 44 Ma for pyroxene (AMDEL Report G7501/88 part 2). The report of the dating laboratory, Australian Minerals Development Laboratories, recommends that, in view of alteration, these ages should be considered "general estimates". The pyroxene ages are in broad agreement with the 1116 ± 12 Ma estimate for Lakeview Dolerite, but the plagioclase age is much younger. A possible reason is that plagioclase is susceptible to hydrothermal solutions even at very low temperatures (Hanes et al., 1986).

Lastly Fig. 10 shows the latitudinal positions of Australia at the time of acquisition of the IM, IC-I and IAR remanences in the Mount Isa Inlier, and coeval remanences in other Australian Precambrian units.

6. Conclusion

Eleven dolerite dykes and five dolerite plugs of Proterozoic age in the Mount Isa Inlier revealed two remanences: the first with a direction close to the present axial dipole field and the second with a vertical downward direction. The first remanence was obtained from four dykes, two basalts, and five dolerite plugs, all of which are metamorphosed. Its resistance to AF and thermal demagnetization suggests that the similarity to the present field is purely coincidental; its presence in dykes of different ages strongly suggests an overprint acquired at the last major peak or in the waning stages of regional metamorphism in the Mount Isa Inlier, i.e., at 1550–1500 Ma, or during subsequent cooling. The second remanence was obtained from seven unmetamorphosed dykes, and is consistent with the IA direction from the 1116 Ma Lakeview Dolerite of Duff and Embleton (1976).

The results agree with the path of Idnurm and Giddings (1988), but require a slight deepening of the 1.5–1.65 Ga loop on that path. The combined pole IAR of 1100 Ma from unmetamorphosed dykes in the Mount Isa Block indicates a fairly high palaeolatitude of 76°. On the other hand, the IM pole of around 1550 Ma suggests the middle palaeolatitude of 30°, close to the present latitude.

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